

POKHARA UNIVERSITY

Level: Bachelor
Programme: BCA

Semester: Spring

Course: Computer Graphics and Multimedia Technology
(Old)

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Computer Graphics? How has the evolution of computer graphics technology impacted the gaming industry in terms of realism and player immersion? 7
b) Define frame buffer. Differentiate between raster and random scan display system. 8
2. a) Describe the DDA Algorithm by calculating necessary derivatives. Compare DDA Algorithm and Bresenham's Line Algorithm. 7
b) How can decision parameter be used to draw circle? Calculate all pixels of a circle in the first quadrant where radius=5 and center at (10, 5). 8
3. a) What do you mean by homogeneous coordinate? Find the composite transformation matrix for anti-clockwise rotation of 45 degree about a point (2,3). Use it to rotate a triangle ABC with vertices A (0,0), B (6,2) and C (3,4). 8

OR

Why do we need clipping? Explain Cohen-Sutherland Line Clipping algorithm with necessary example.

- b) Differentiate between 2D and 3D graphics? In graphics which dimensional is more applicant and explain why? 7
4. a) Define projection. Derive the expression and matrix representation for oblique parallel projection. 8
b) Explain the importance of shading method. Compare and contrast between Gouraud and Phong shading model. 7
5. a) What is multimedia technology? Explain about the multimedia teams. 8
b) What is Multimedia Authoring? List different Multimedia Authoring tools and explain any two. 7

6. a) Explain RGB and CMYK color model, Why CMYK is called subtractive color and RGB is called additive color model? 8
- b) What is Animation? Explain the principles of Animation for today's development. 7
7. Write short notes on: (**Any two**) 2×5
- a) LCD vs LED
- b) Graphics and Image Processing
- c) OpenGL